DEVOXX MOROCCO

Why Your RAG Agent Is Confidently Wrong

Retrieval Choices That Actually Matter

David vonThenen
AI/ML Engineer
@davidvonthenen



David vonThenen

- Are you Human or an AI?
- I want 5 Kubernetes
- Virtual Machines are Real
- Cloudy, cloudy, cloudy...
- There is storage for that!

@davidvonthenen



Agenda

- The "R" In RAG: Exploring Methods
  - Vector, BM25, and Graph Retrieval
- Discussions on Data Retrieval Challenges
- What is Hybrid RAG? Why Use It?
  - BM25 + Vector Embeddings
  - Knowledge Graphs + Vector Embeddings
- Fact-based Answers and AI Governance??
- Q & A, Open Discussion, Wrap Up

The ”R” In RAG:

Exploring Retrieval Options

Breakdown of RAG

Training Data Has Cutoff Date

Retrieval

- Provide Data Beyond Cutoff
- Data Retrieval via "A Method"

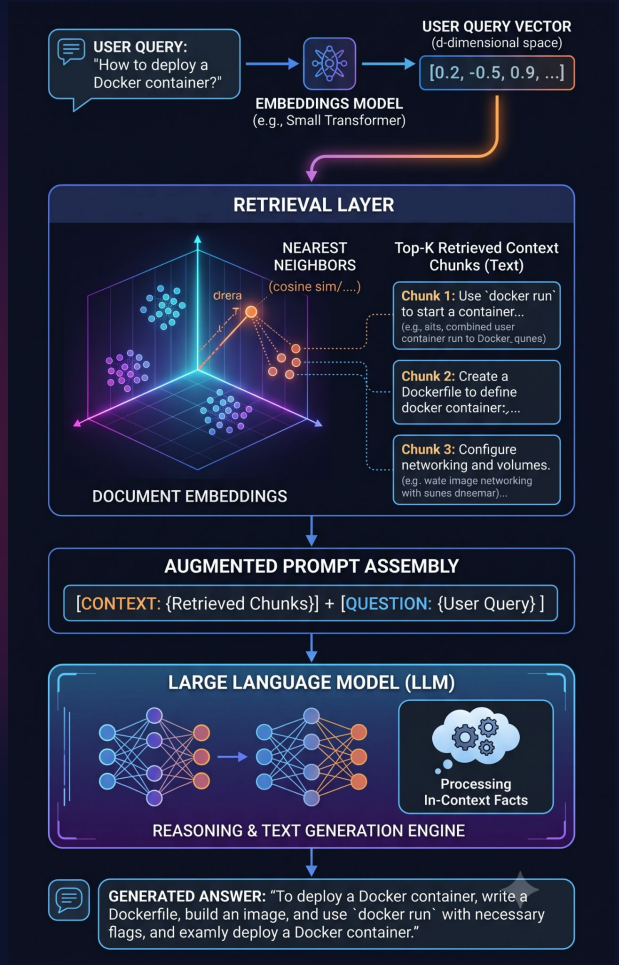
Augmented

- Prompt Assembly
- Provides Context
- Retrieved Data is Truth / Fact

Generation Phase

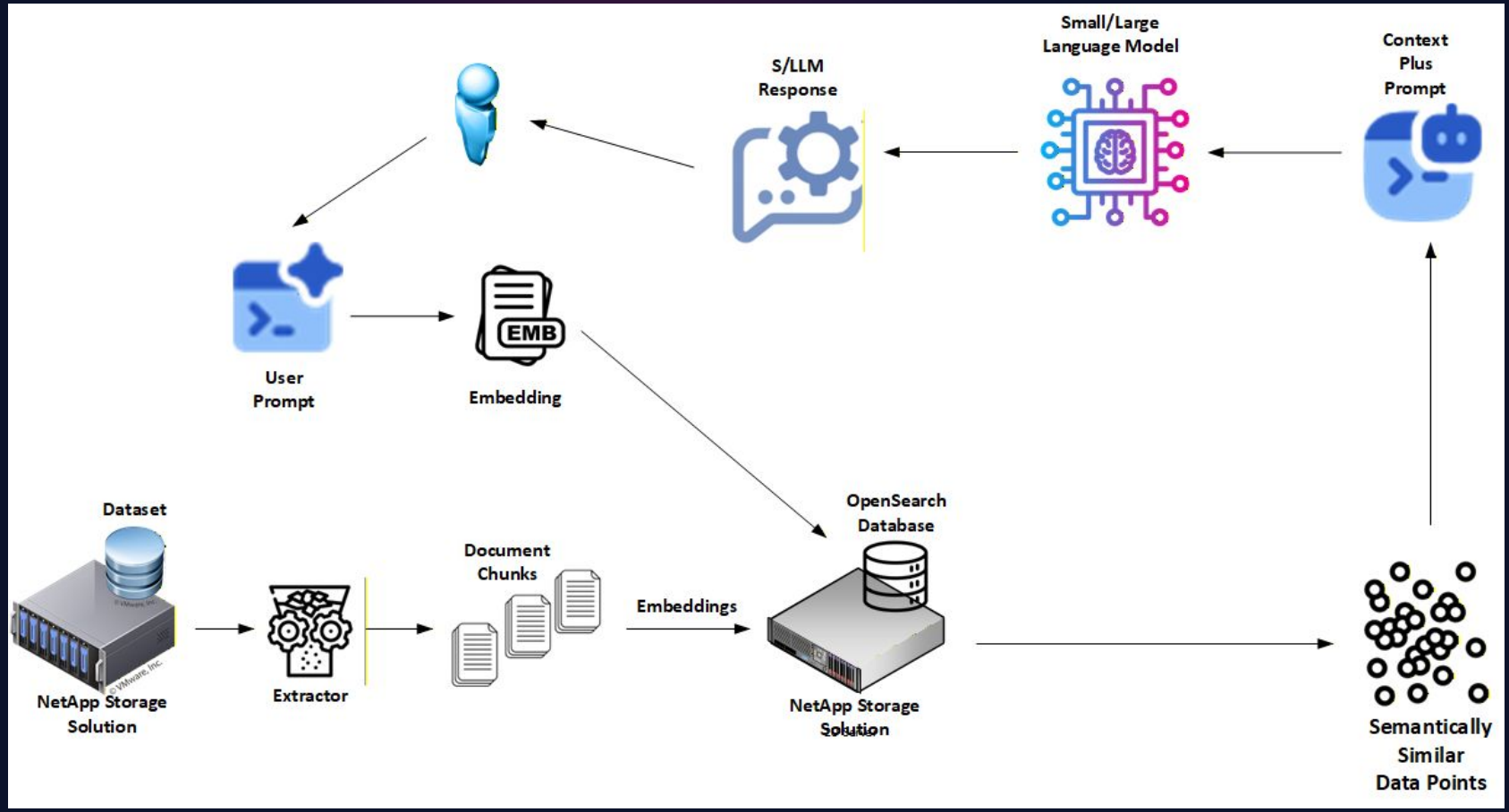
- Lang. Model Synthesizes an Answer

Image Attribution
Gemini Generated



Vector-based RAG

Image Attribution
A David Original



What is Semantic Similarity, Really?

Convert Text to Objects/Vectors:

- Similar Numerical Representation
- Captures Relationships Between:
  - Words, Concepts, etc

Is The Foundation of Vector Search

- Query Nearest Vectors (kNN), like:
  - Cosine Similarity
  - Euclidean Distance

How Does This Relate To Answers?

  - Does Similarity Mean Answers?

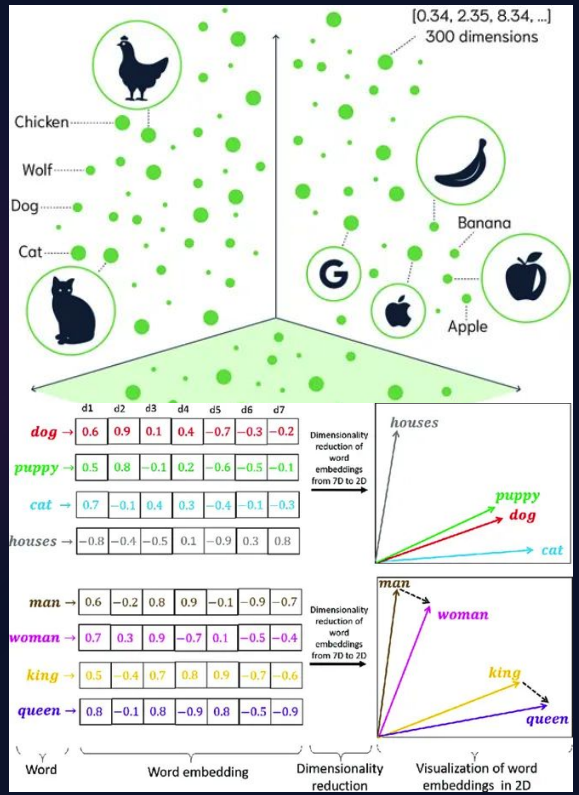
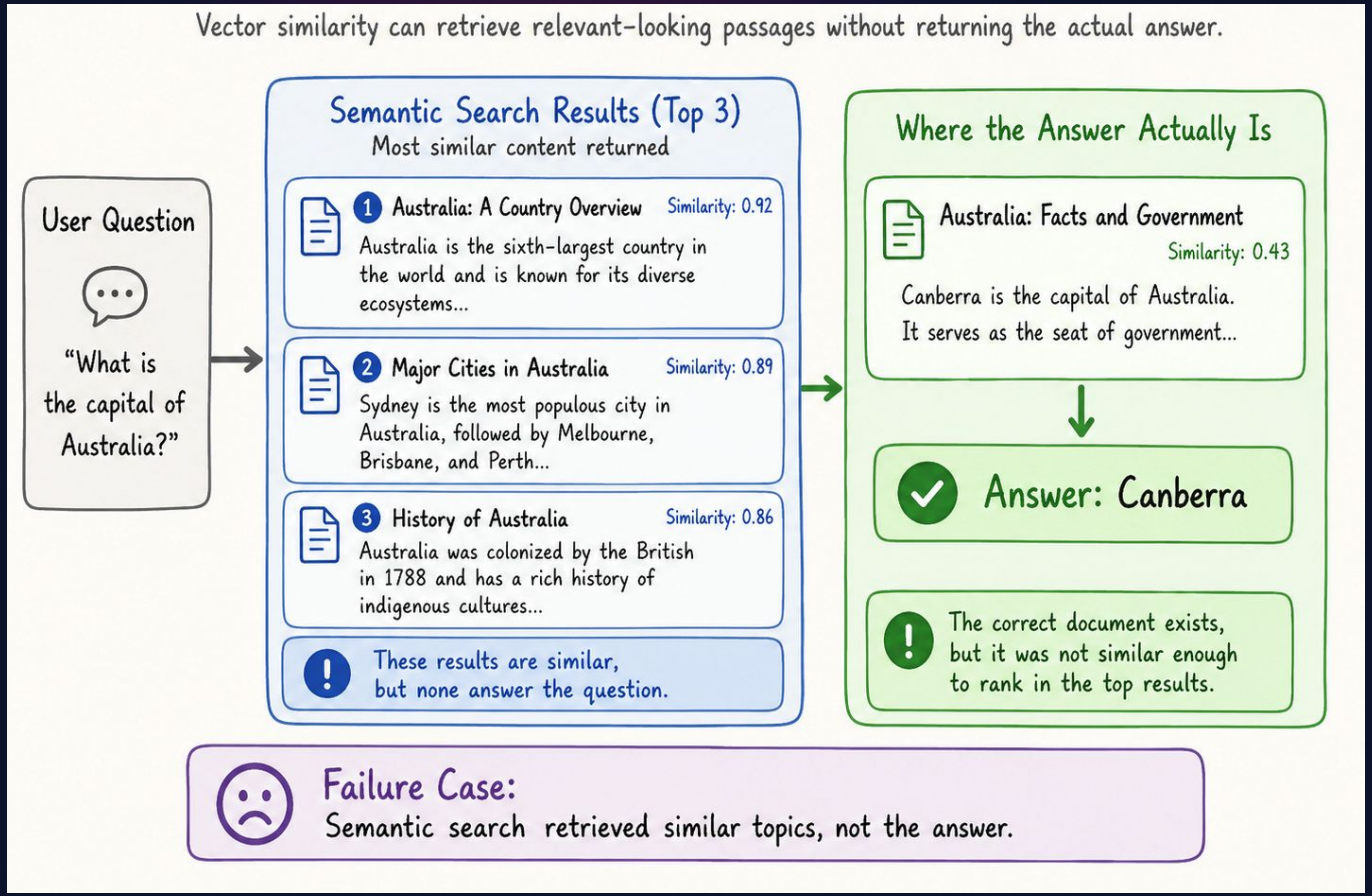


Image Attribution
https://mehmetozkaya.medium.com/understanding-semantic-meaning-and-similarity-search-cosine-similarity-and-euclidean-distance-666fcb5911ea
https://fastdatascience.com/natural-language-processing/semantic-similarity-with-sentence-embeddings/

Similar Context ≠ Answers

Image Attribution
ChatGPT Generated



Context Is Important!

- Conceptually, What Are Answers?
  - Factual Replies To Questions
  - Exact Solution To A Problem
    - Example: Math Problem
- Opinions, Preferences, Guesses are:
  - Answers Influenced By:
    - Experience, Intuition, Emotion
  - When No Answer Is Known
    - Example: Conjecture/Postulate
- Now, Think About Similar Data Points

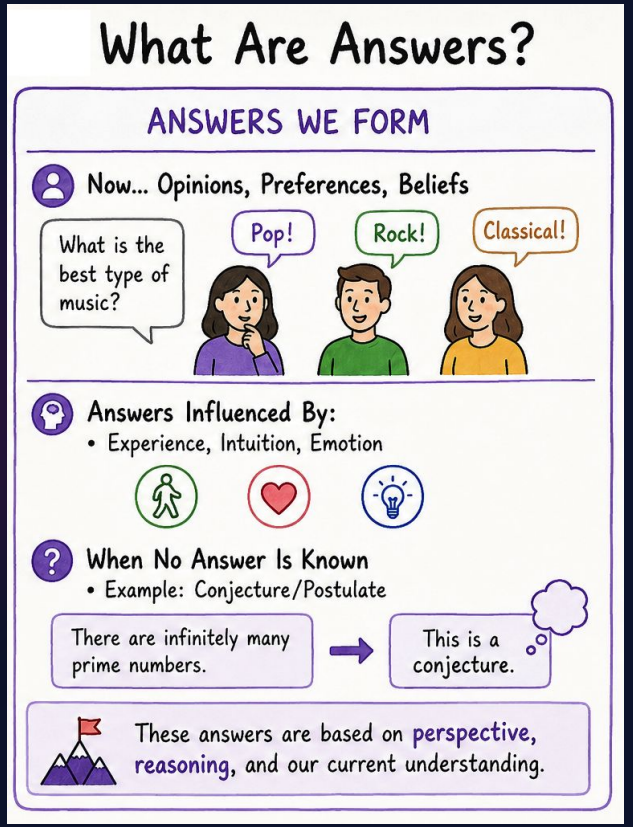
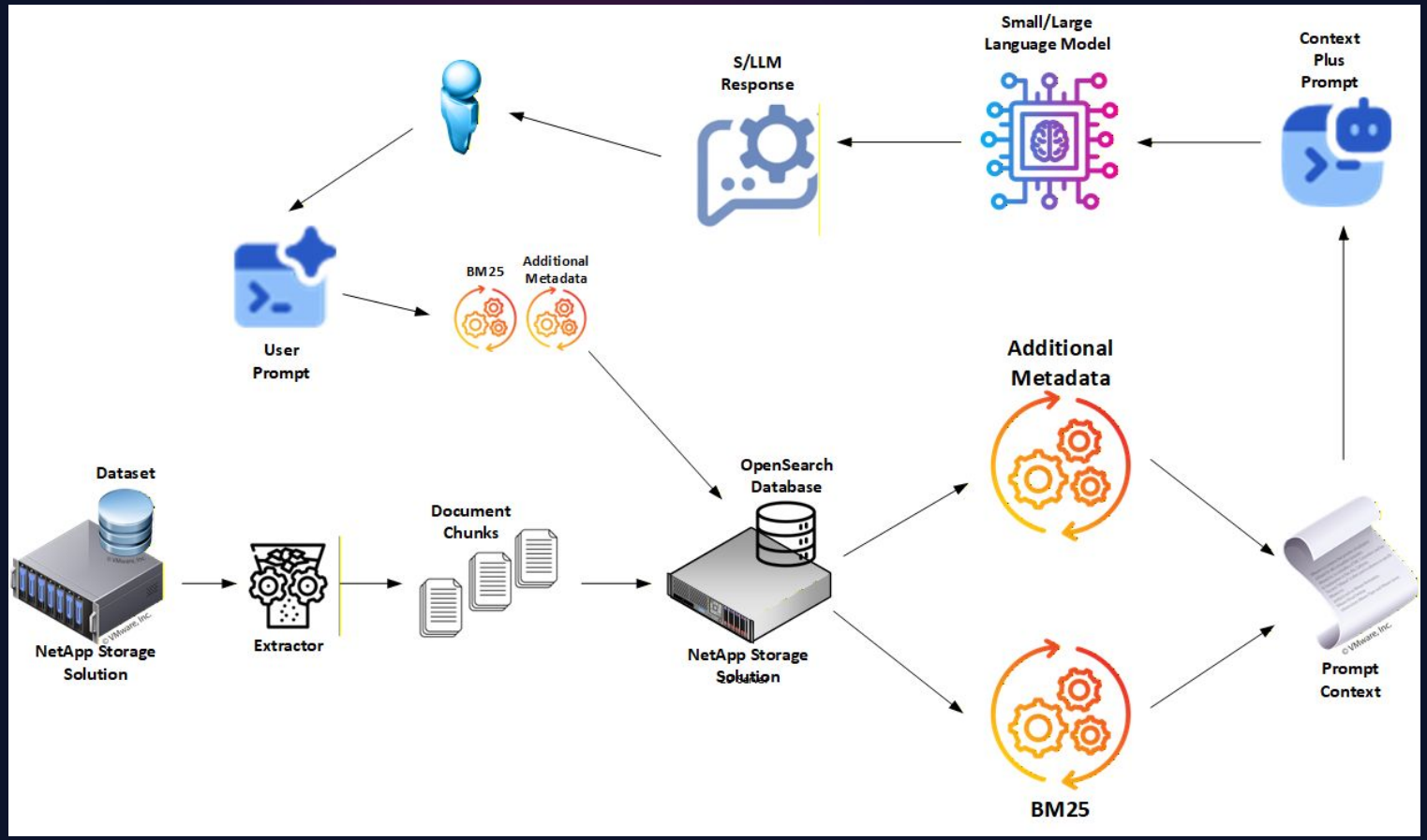


Image Attribution
ChatGPT Generated

BM25-based RAG

Image Attribution
A David's Original



BM25-based Retrieval

- Best Match Based on Keywords
  - Rank Corpus on Token Appearance
  - Deterministic Identifier Matching
  - Scores Document Relevance
- Retrieval Phase
  - Key (BM25) / Value (Chunk)
  - Push More Metadata In Keys
  - Compounded Queries
- Augmentation Phase
  - Near Identical to Vector Embed.

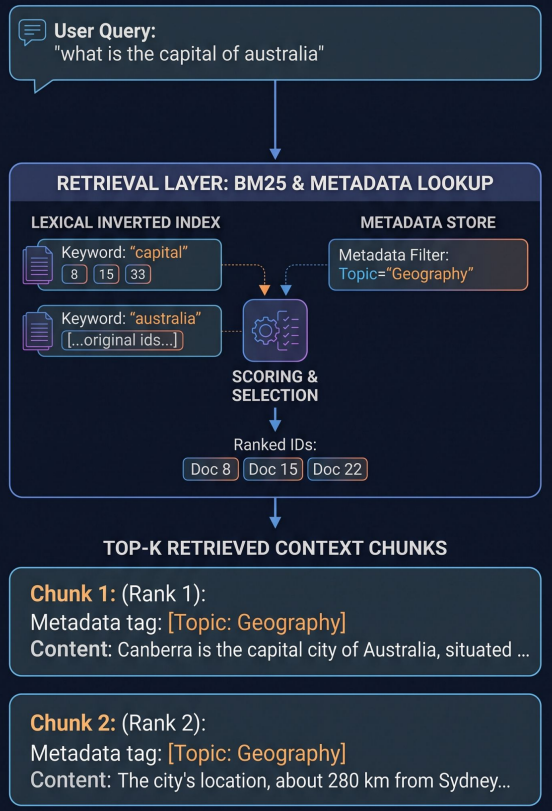


Image Credit:
Gemini Generated

Named Entity Recognition

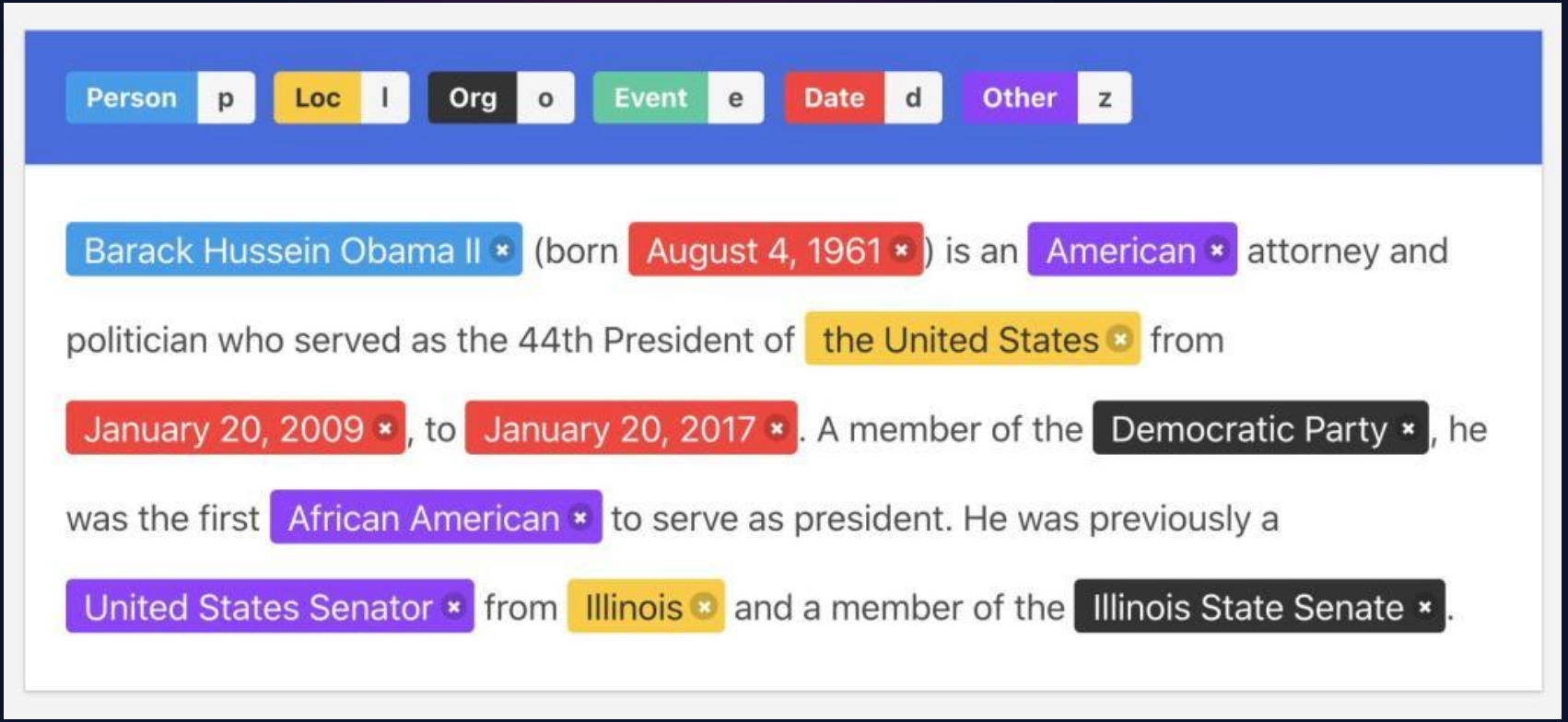


Image Credit:
A Beginner's Introduction to NER (Named Entity Recognition) by Adwait
https://www.analyticsvidhya.com/blog/2021/11/a-beginners-introduction-to-ner-named-entity-recognition/

Semantic vs BM25 Retrieval

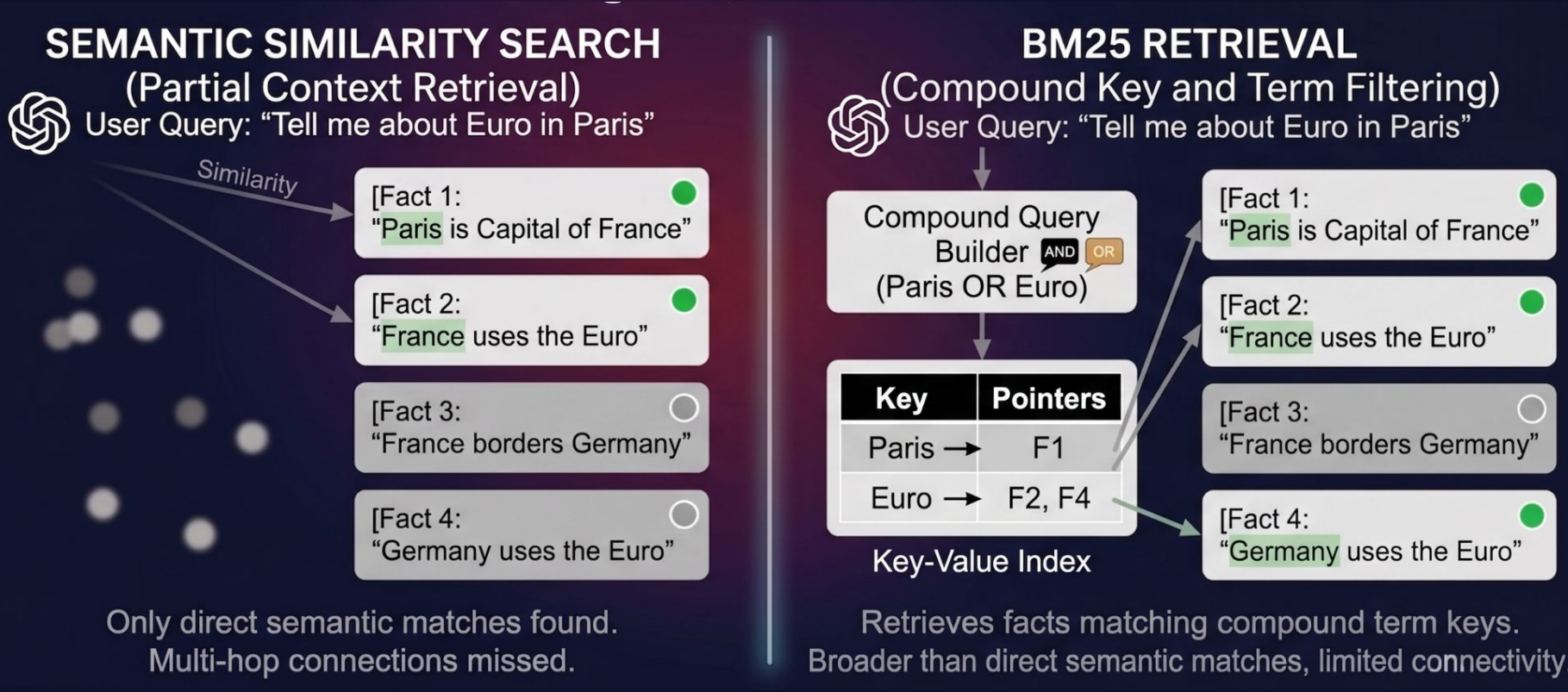
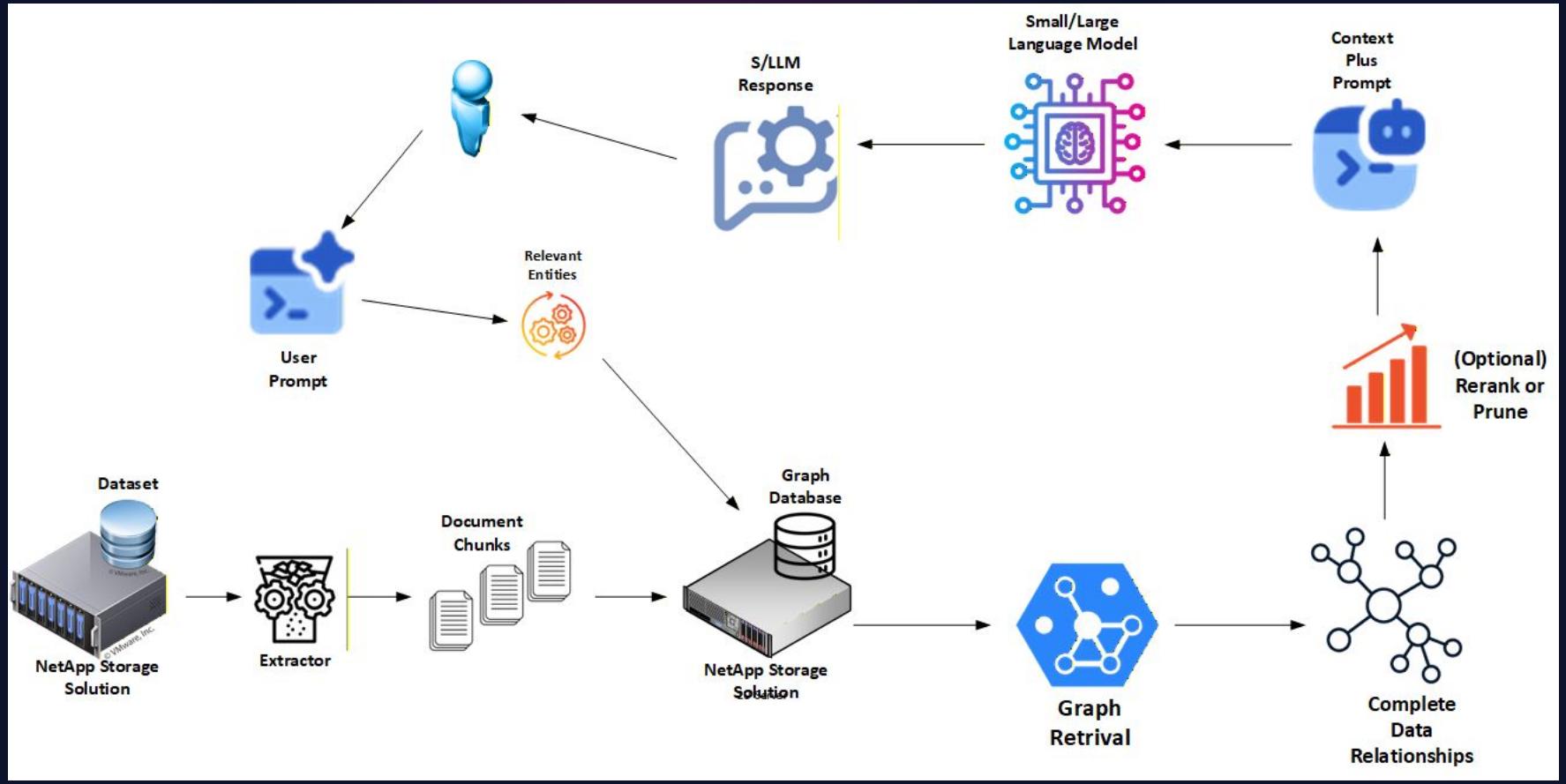


Image Credit:
Gemini Generated

Graph-based RAG

Image Attribution
A David's Original



Graph-based Retrieval

- What Are Knowledge Graphs?
  - Retrieval Based on Data Relationships
    - Association Between Data Points
  - Retrieve a Network of Facts
- Benefits:
  - Highest Retrieval Relevance
  - Most Accurate Collection of Facts
  - Best Answers for Users
- Drawbacks:
  - Requires Structured Data
  - Manage Ontologies (Like, in SQL)

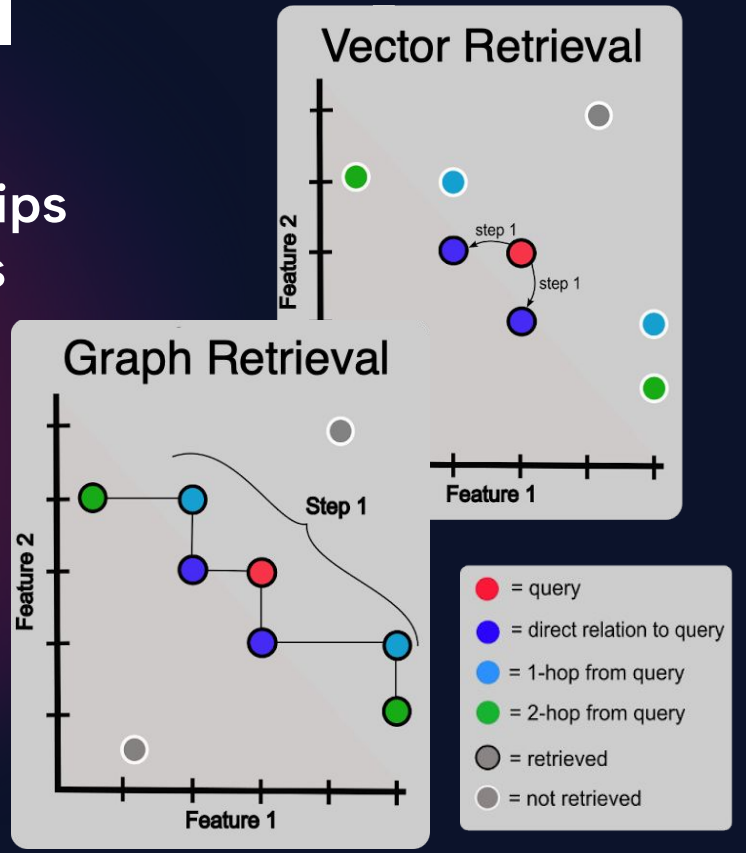


Image Credit:
Nick Alonso, Beren Millidge
Understanding Graph-based RAG and Multi-Hop Question Answering

Semantic vs Graph Retrieval

SEMANTIC SIMILARITY SEARCH
(Partial Context Retrieval)

User Query: “Tell me about Euro in Paris”

Similarity

[Fact 1:
“Paris is Capital of France”

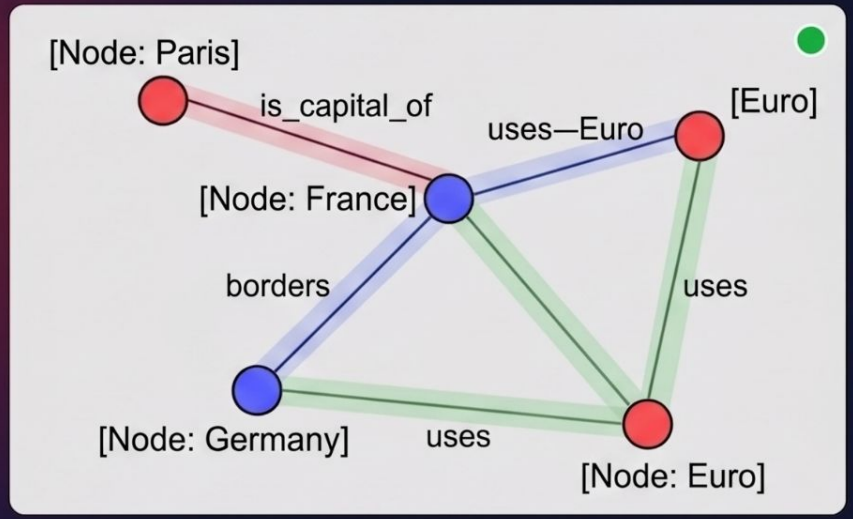
[Fact 2:
“France uses the Euro”

[Fact 3:
“France borders Germany”

[Fact 4:
“Germany uses the Euro”

Only direct semantic matches found.
Multi-hop connections missed.

KNOWLEDGE GRAPH RETRIEVAL
(Comprehensive Network Retrieval)



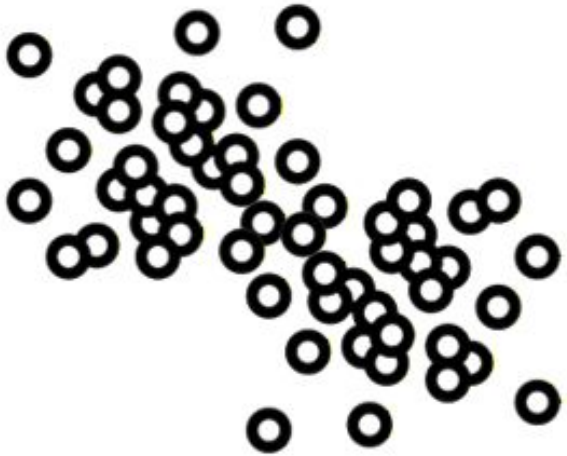
All associated facts, including
multi-hop relationships, retrieved.

Image Credit:
Gemini Generated

From the Data Point Perspective

Image Attribution
A David's Original

Vector Embeddings



K-Nearest Neighbor (KNN)

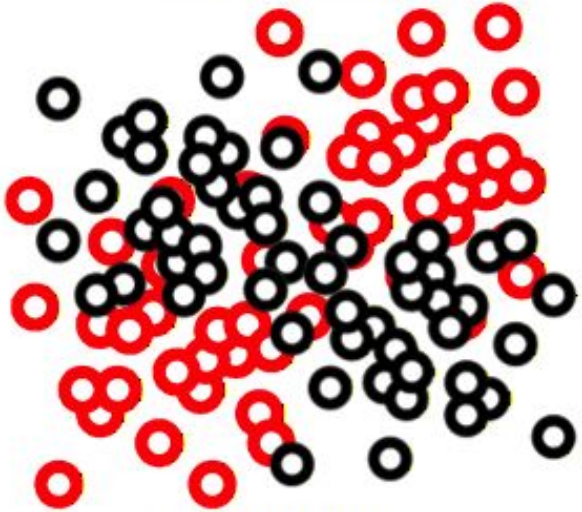
Knowledge Graph

BM25

From the Data Point Perspective

Image Attribution
A David's Original

Vector Embeddings



Approximate Nearest Neighbor (ANN)

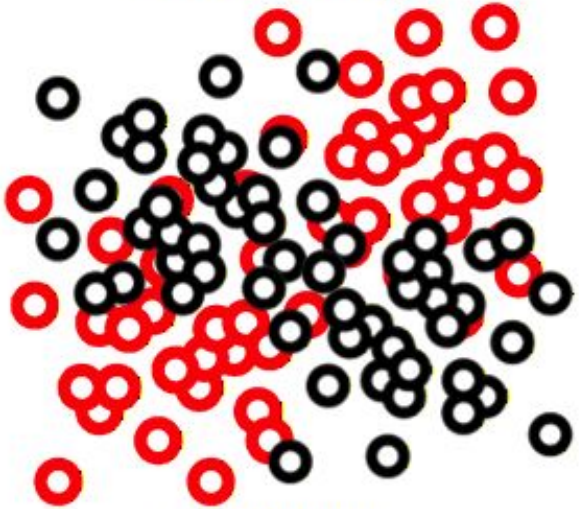
Knowledge Graph

BM25

From the Data Point Perspective

Image Attribution
A David's Original

Vector Embeddings



Approximate Nearest Neighbor (ANN)

Knowledge Graph

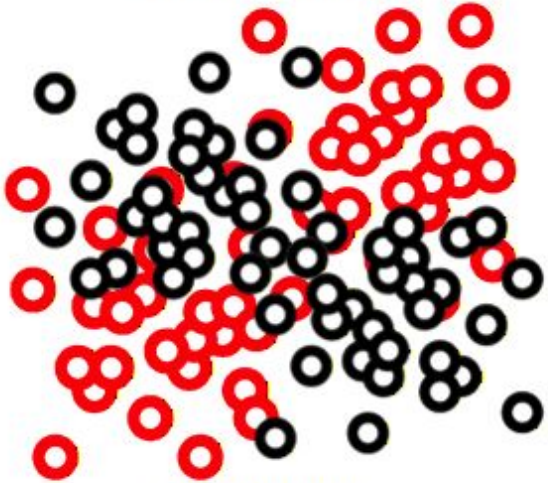


BM25

From the Data Point Perspective

Image Attribution
A David's Original

Vector Embeddings

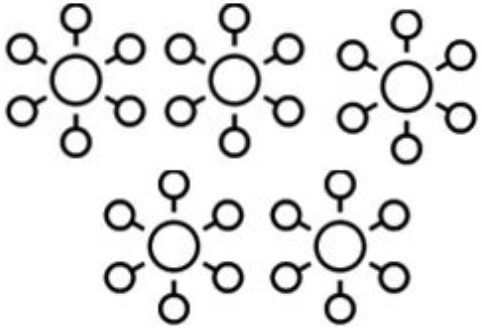


Approximate Nearest Neighbor (ANN)

Knowledge Graph

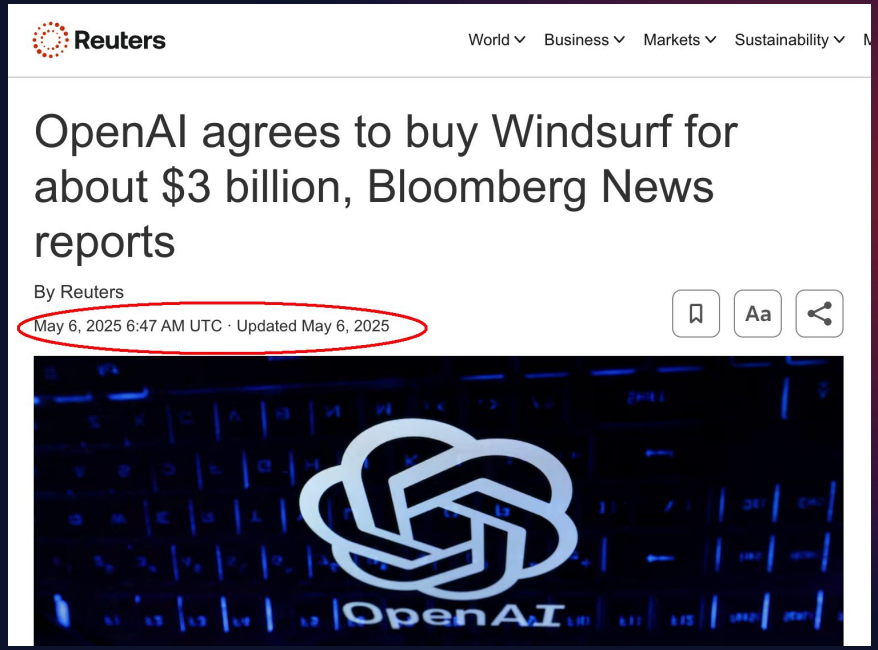


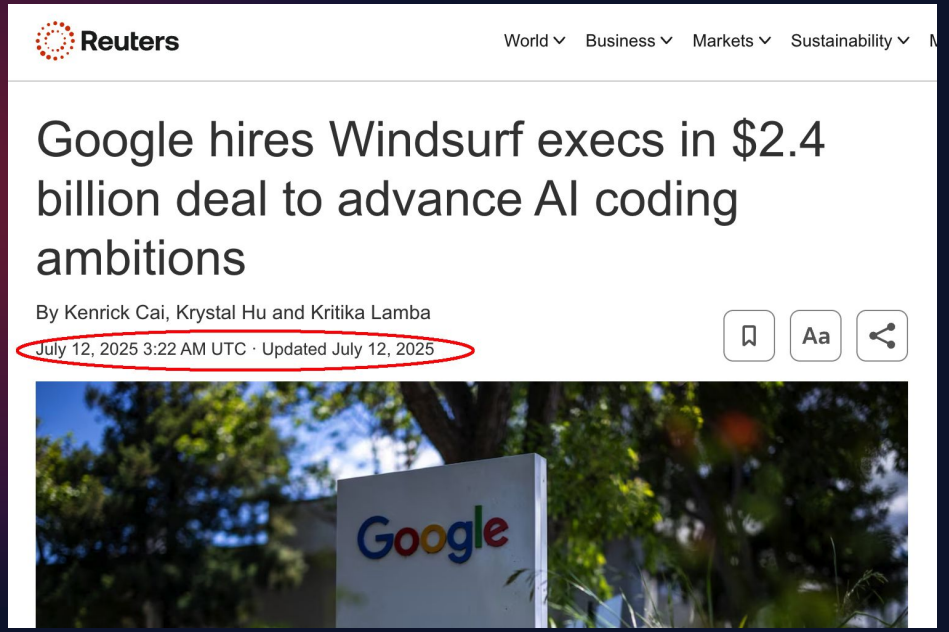
BM25



Three Methods, Three Answers

Let's Look At Two Different Realities...





Section 1: Different RAG Agents

github.com/davidvonthenen/2026-devoxx-morocco

Task 1: Simple Vector-based Agent

Task 2: Simple BM25-based Agent

Task 3: Simple Graph-based Agent

Section 1: Results Discussion

Take a Look at History

- May 06, 2025: OpenAI to Acquire Windsurf
- July 12, 2025: Google Acqui-hired for $2.4 billion

Results: Vector 🤔, BM25 😅, Graph 😅

Takeaways:

- MINIMAL Vector, BM25, Graph Implementations
- Implementing BM25 and Graph Doesn't Yield Magic!
  - Some Vector Workarounds (Square Peg, Round Hole)
- Data Retrieval (Ranking and Conflict Resolution) are the KEY!

Image Attribution
Princeton University Researcher - Tool helps clear biases from computer vision
Molly Sharlach on October 1, 2020

Data Retrieval Challenges

Discussions on Finding Answers

Let's Take a Concrete Example

- Build a "Simple" Recipe Chatbot for Cooking
  - For Dinner Tonight, I Want to Cook...
- Use Cases:
  - Search Recipes Based Likes and Dislikes
  - Ask About Recipe Ingredients
  - Exclude Allergens (Soy, Dairy, etc)
    - High Stakes: I Will Die If I Eat a Peanut
  - Ask Questions On Preparing the Recipe
- Simple Chatbot, Serious Consequences



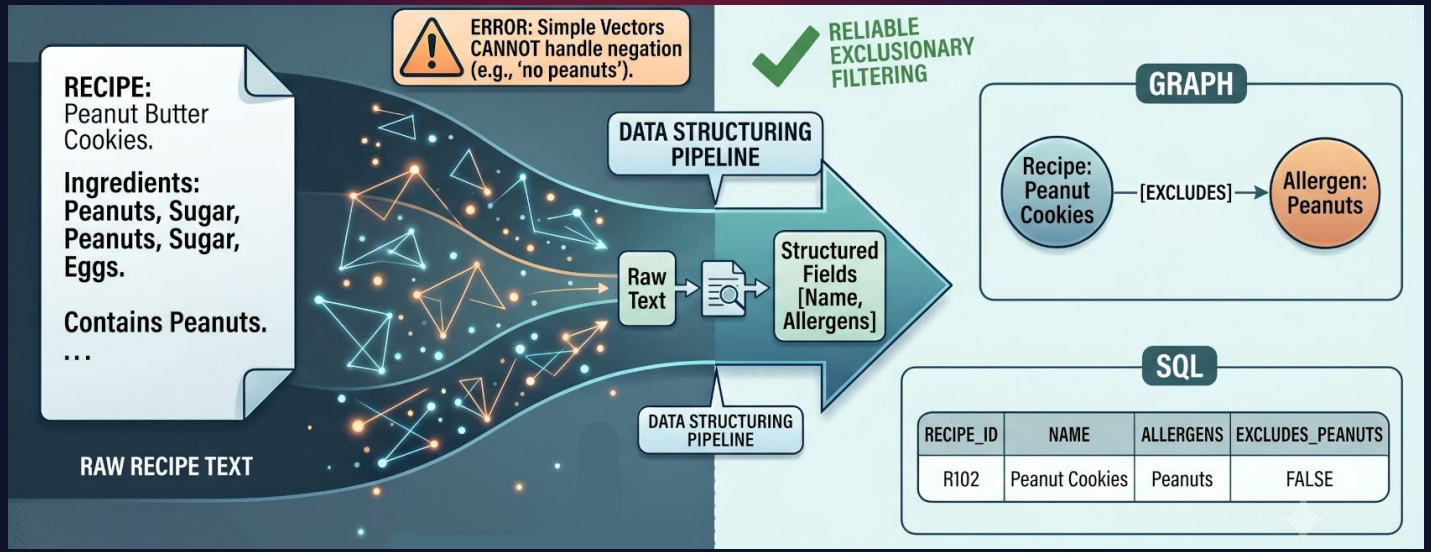
Image Attribution
Gemini Generated

Vector Possibilities

Image Attribution
Gemini Generated

Very CPU Intensive or Expensive GPU:

- Unstructured –> Filtering via Tags (Doesn't Scale Well)
  - OpenSearch (sparse search), Milvus (Scalar Filtering), Quadrant (must_not)
- OpenSearch: Inverted BM25 index



Structured Retrieval is a Certainty!

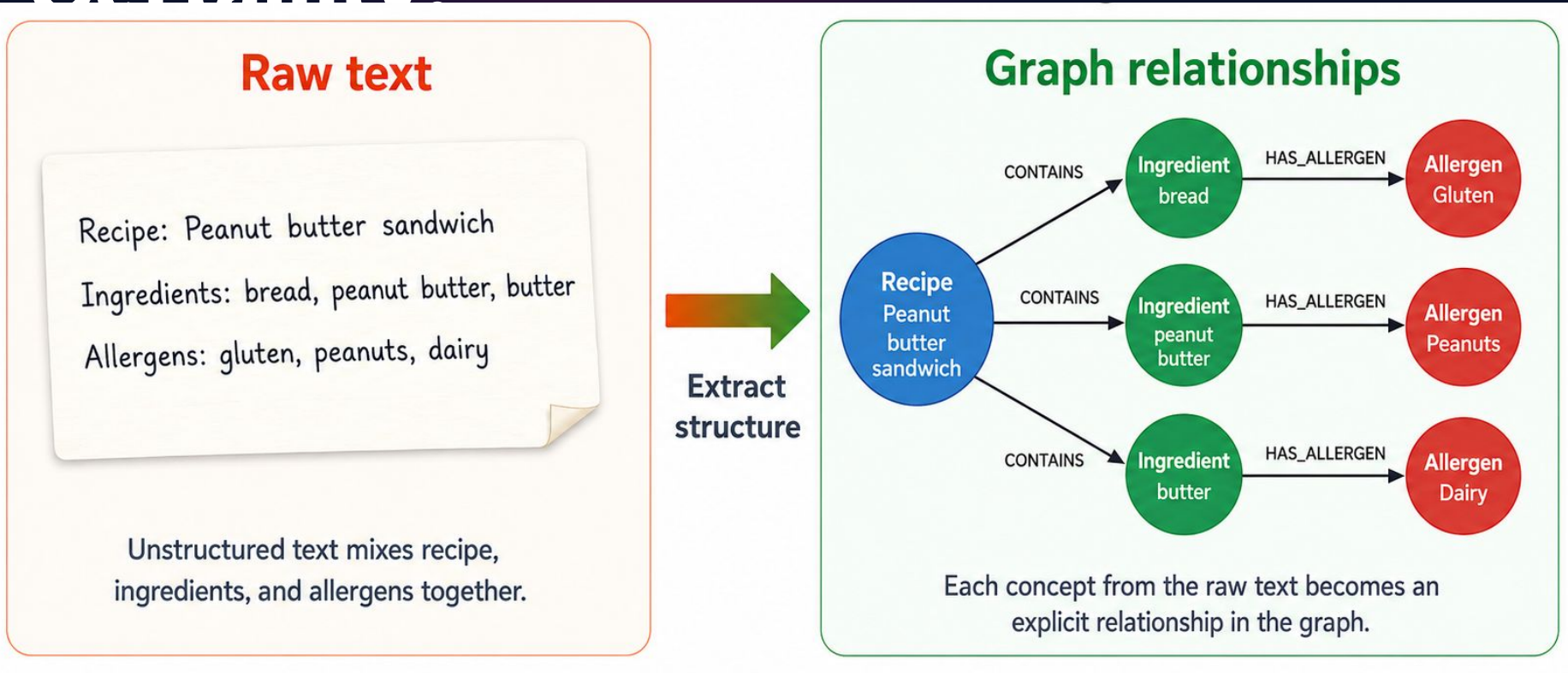


Image Attribution
ChatGPT Generated

Again… Consider Alternatives!

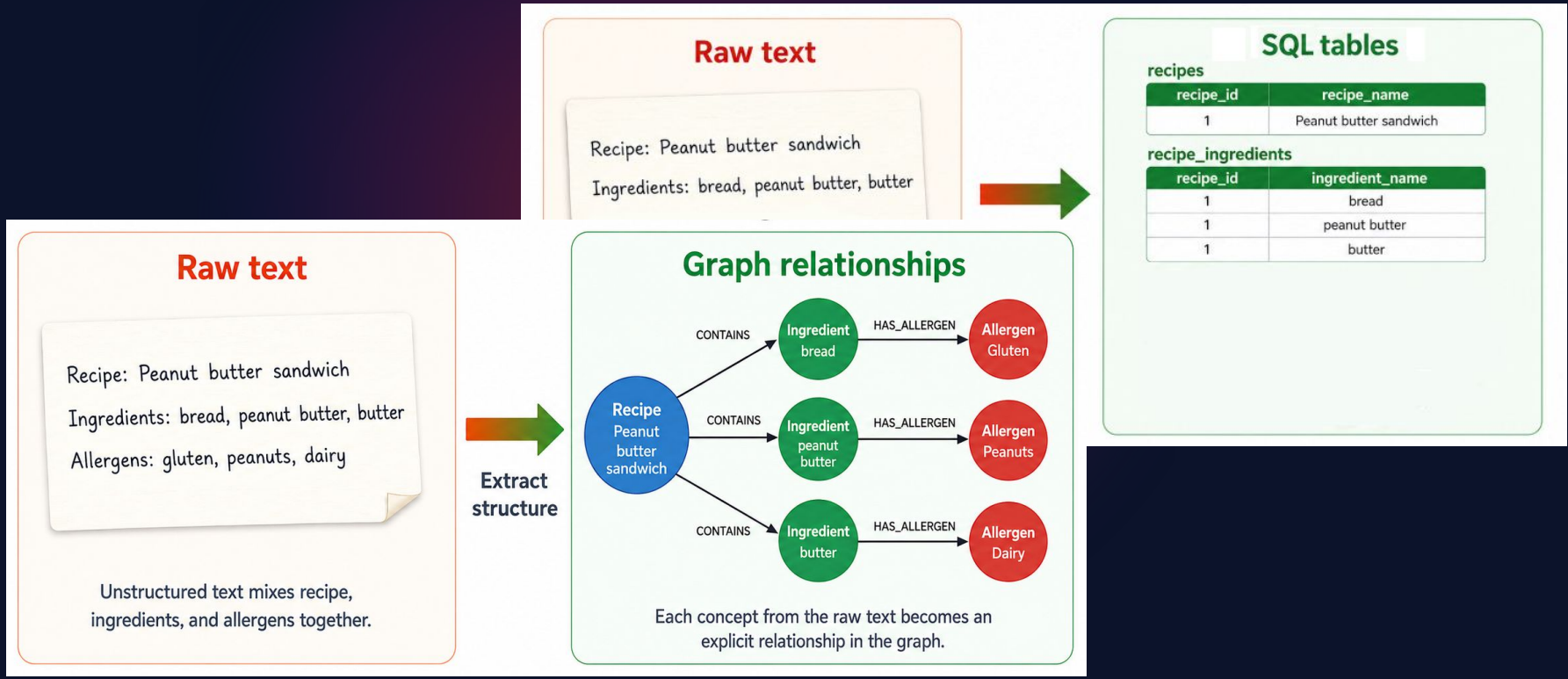
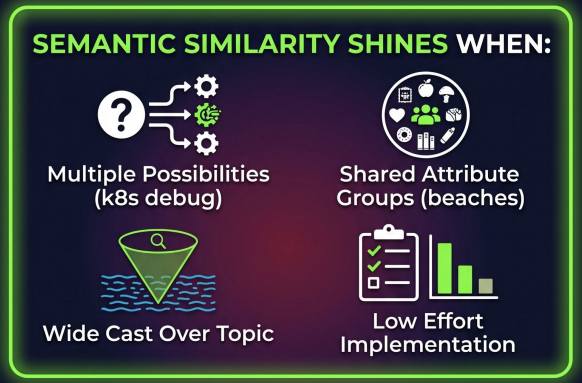


Image Attribution
ChatGPT Generated

Pros & Cons: Unstructured Data

- Semantic Similarity Shines When:
  - There Isn't a Single Answer
  - Want to Try Possibilities
    - Debugging -> What to Try
    - "k8s pod in crash loop"
  - Want a Collection of "Things"
    - Shared Attributes in Groups
    - "Beaches Having High Salt Content"
  - Want to Cast a Wider Net In 1 Topic
  - Easy to Implement, Low Effort
- Cons: The Elephant in the Room...



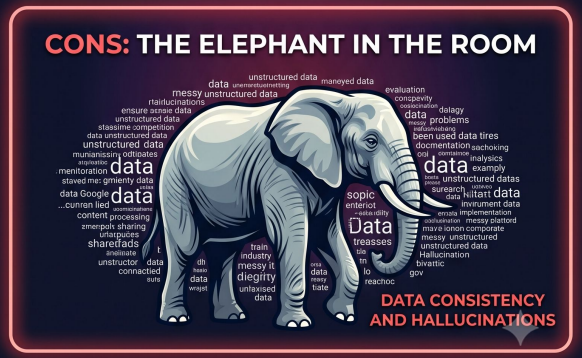


Image Attribution
Gemini Generated

Pros & Cons: Structured Data

- Structured Retrieval Shines When:
  - Answers Are Absolute or Singular
    - Example: Math Problem
  - Has a Generally "Accepted" Answer
    - Example: The Sky is Blue
      - Can be Red, Smokey, etc
  - Highly Regulated Environments
  - Requirement for Factual Answers
- Cons:
  - Unstructured ➡️ Effort Is Non-Trivial

Image Attribution
Gemini Generated

WHERE IT SHINES (PROS)


Absolute/Singular Answers


Generally “Accepted” Facts


Highly Regulated Environments



Non-Negotiable Facts

CONS (THE EFFORT)



Unstructured Data


Non-Trivial Effort to Prepare

If You Structure A Lot of the Data...

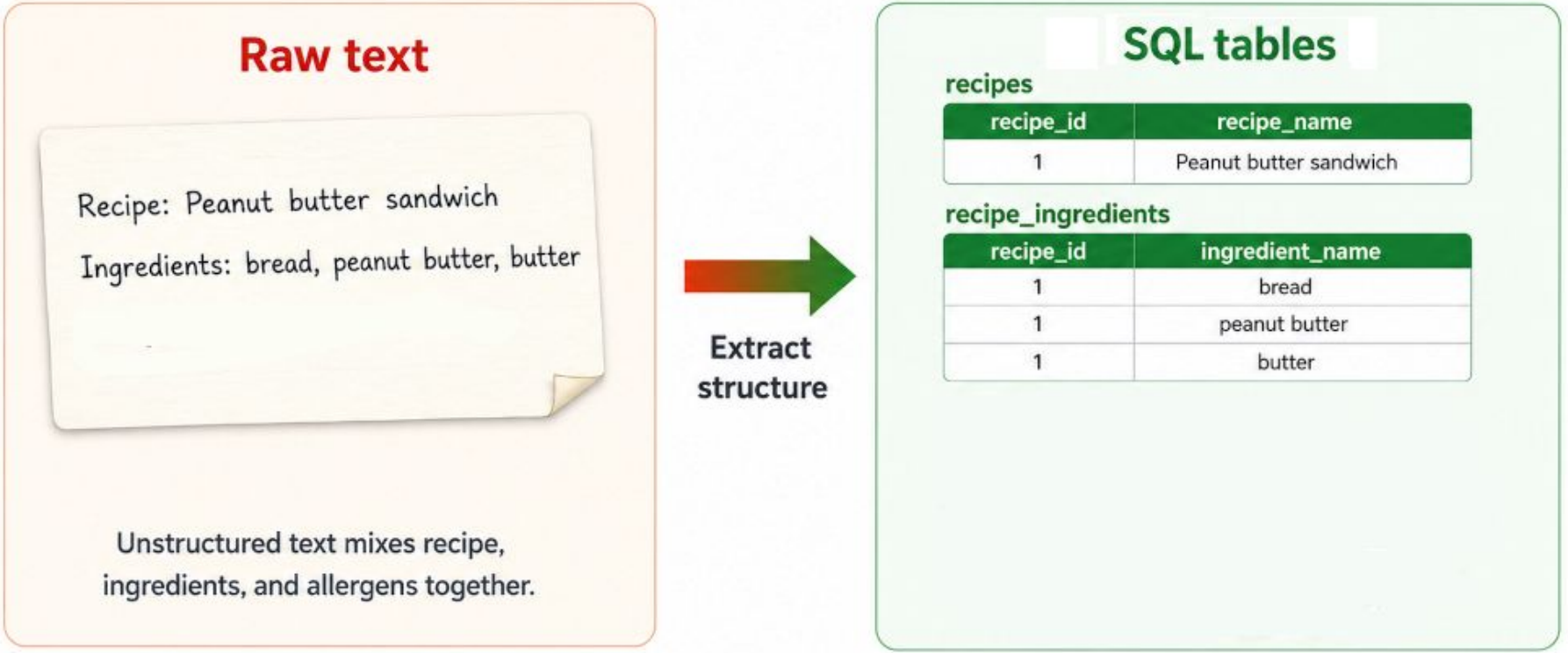


Image Attribution
ChatGPT Generated

Why Not Structure All Of It!

Raw text

Recipe: Peanut butter sandwich

Ingredients: bread, peanut butter, butter

Allergens: gluten, peanuts, dairy

Unstructured text mixes recipe, ingredients, and allergens together.

Extract structure

SQL tables

recipes

recipe_id	recipe_name
1	Peanut butter sandwich

recipe_ingredients

recipe_id	ingredient_name
1	bread
1	peanut butter
1	butter

recipe_allergens

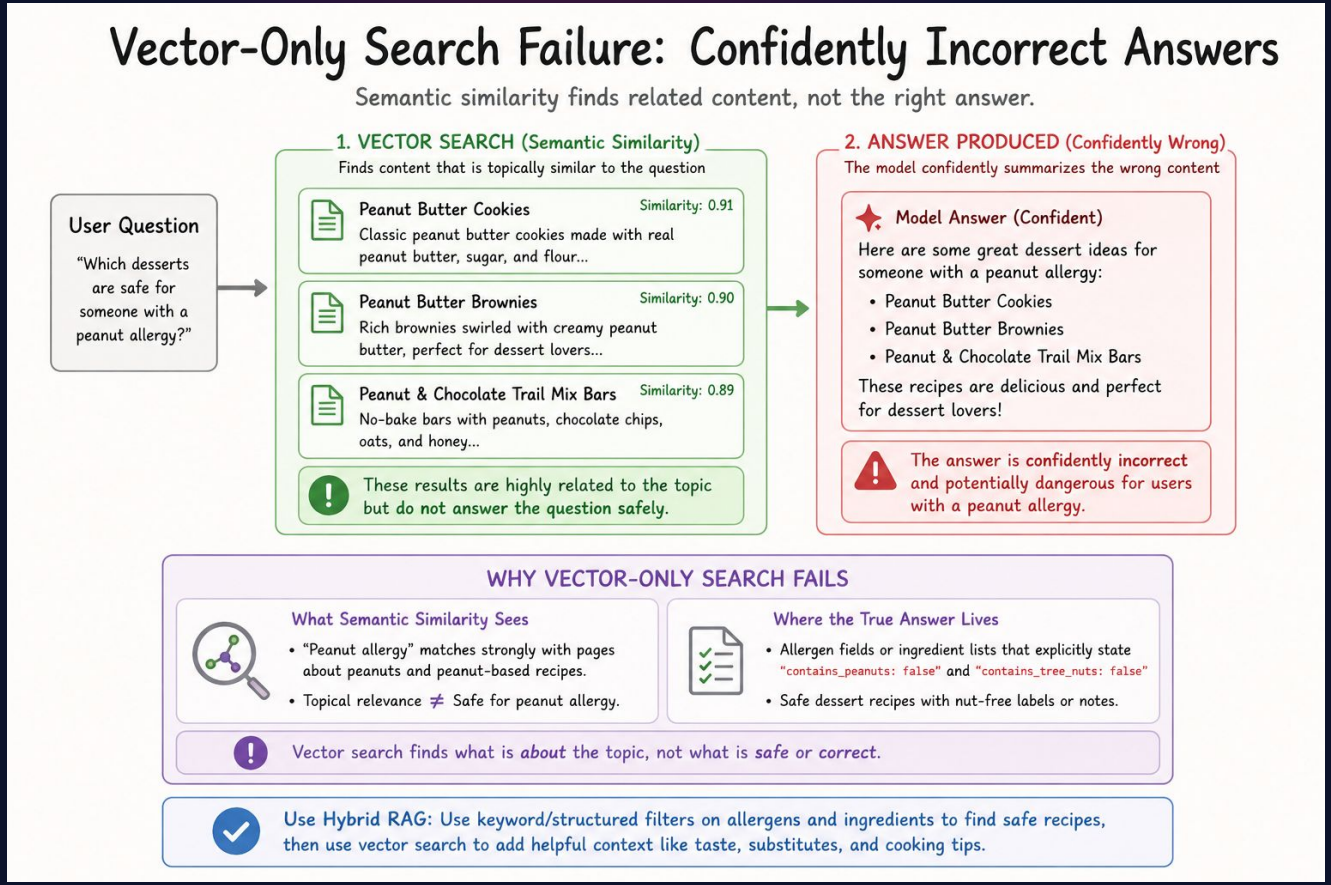
recipe_id	allergen_name
1	Gluten
1	Peanuts
1	Dairy

Every concept from the raw text is separated into queryable columns and rows.

Image Attribution
ChatGPT Generated

Similar Doesn't Mean Facts!

Image Attribution
ChatGPT Generated



Section 2: Retrieval Mechanisms

github.com/davidvonthenen/2026-devoxx-morocco

Task 1: Exploring Vector Retrieval

Task 2: Exploring BM25 Retrieval

Task 3: Exploring Graph Retrieval

Section 2: Results Discussion

Other Data Architectural and Retrieval Limitations:

- Geometric Ceiling of Single-Vector Spaces
- Token Dilution, Polysemy, and Exact Identifier Erasure
- Context Blindness and Structural Segmentation
- Temporal Collapse and Static Latent Space

Results: Vector 😭, BM25 😅, Graph 🥳

Takeaways:

- The More Structure to Your Data, Better Possibilities
- Structure is Great, Still Need to Retrieve It!

Image Attribution
Princeton University Researcher - Tool helps clear biases from computer vision
Molly Sharlach on October 1, 2020

What Is Hybrid RAG?

More Importantly, Why Should I Use It?

Hybrid RAG

Image Attribution
https://arxiv.org/html/2408.04948v1
arXiv:2408.04948v1 [cs.CL] 09 Aug 2024

Cornell University

We gratefully acknowledge support from the Simons Foundation, member institutions, and all contributors.

Donate

arXiv > cs > arXiv:2408.04948

Search... All fields Search

Help | Advanced Search

Computer Science > Computation and Language

[Submitted on 9 Aug 2024]

HybridRAG: Integrating Knowledge Graphs and Vector Retrieval Augmented Generation for Efficient Information Extraction

Bhaskarjit Sarmah, Benika Hall, Rohan Rao, Sunil Patel, Stefano Pasquali, Dhagash Mehta

Extraction and interpretation of intricate information from unstructured text data arising in financial applications, such as earnings call transcripts, present substantial challenges

Access Paper:
View PDF
HTML (experimental)
TeX Source
view license

Current browse context:
cs.CL
< prev | next >
new | recent | 2024-08

Change to browse by:
cs
cs.LG
q-fin

What is Hybrid RAG?

- Composed of Two Phased Approach
  - Phase 1: Truth Grounding Phase
    - Use Deterministic Data Retrieval
    - Anchor the Answer to Facts
  - Phase 2: Add Semantic Nuance
    - Use Semantic Similarity for "Flavor"
    - Make the Answer Human-like
- For Each Phase:
  - Retrieve Data Based On The Goal
  - Use a Language Model for Synthesis
- Provide the Answer!

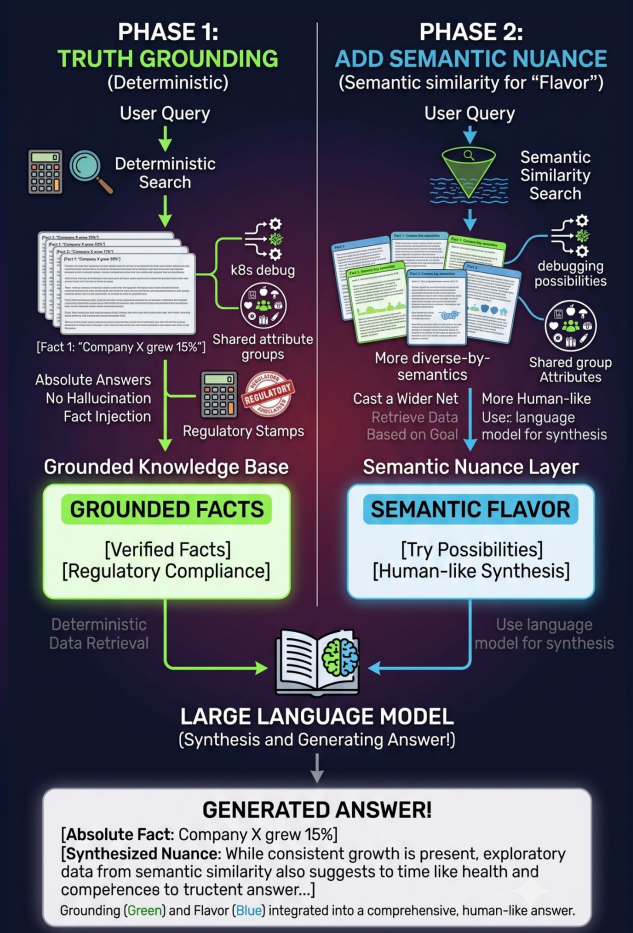
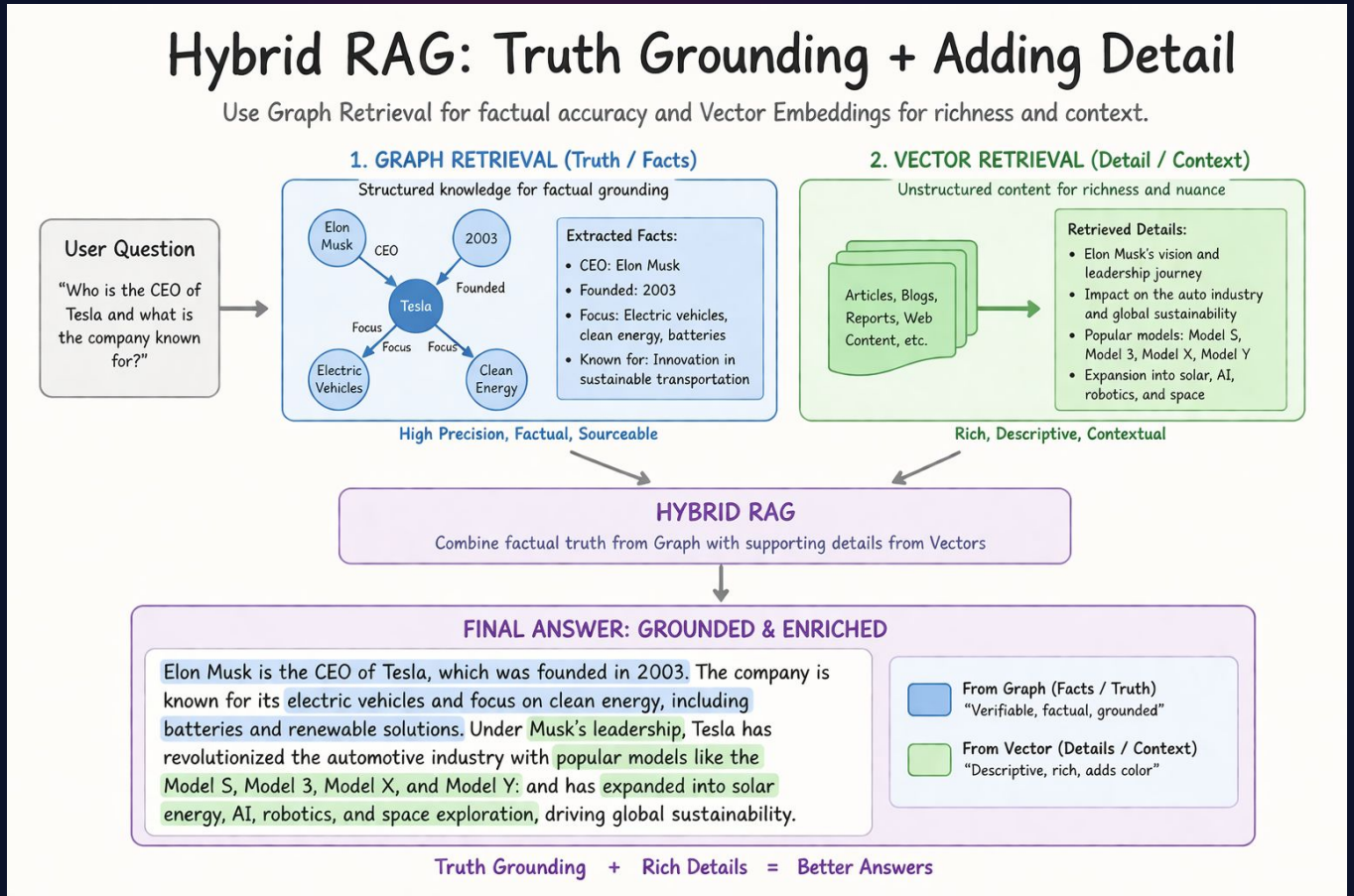


Image Attribution
Gemini Generated

Hybrid RAG Response

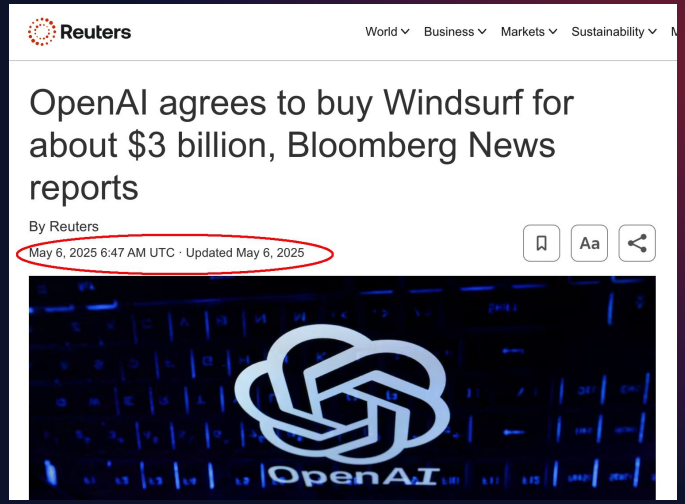
Image Attribution
ChatGPT Generated

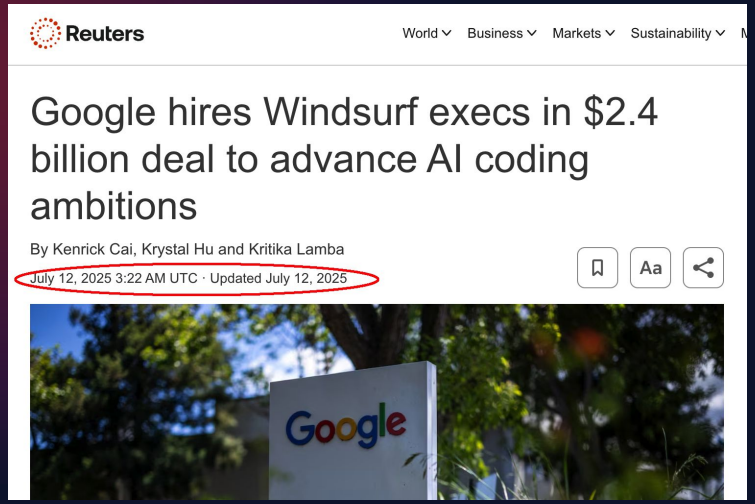


Hybrid RAG References

Reference Architectures (for Unstructured Data):

- github.com/davidvonthenen/hybrid-rag-bm25-with-ai-governance
- github.com/davidvonthenen/hybrid-rag-graph-with-ai-governance





Hybrid RAG

Graph and Vector

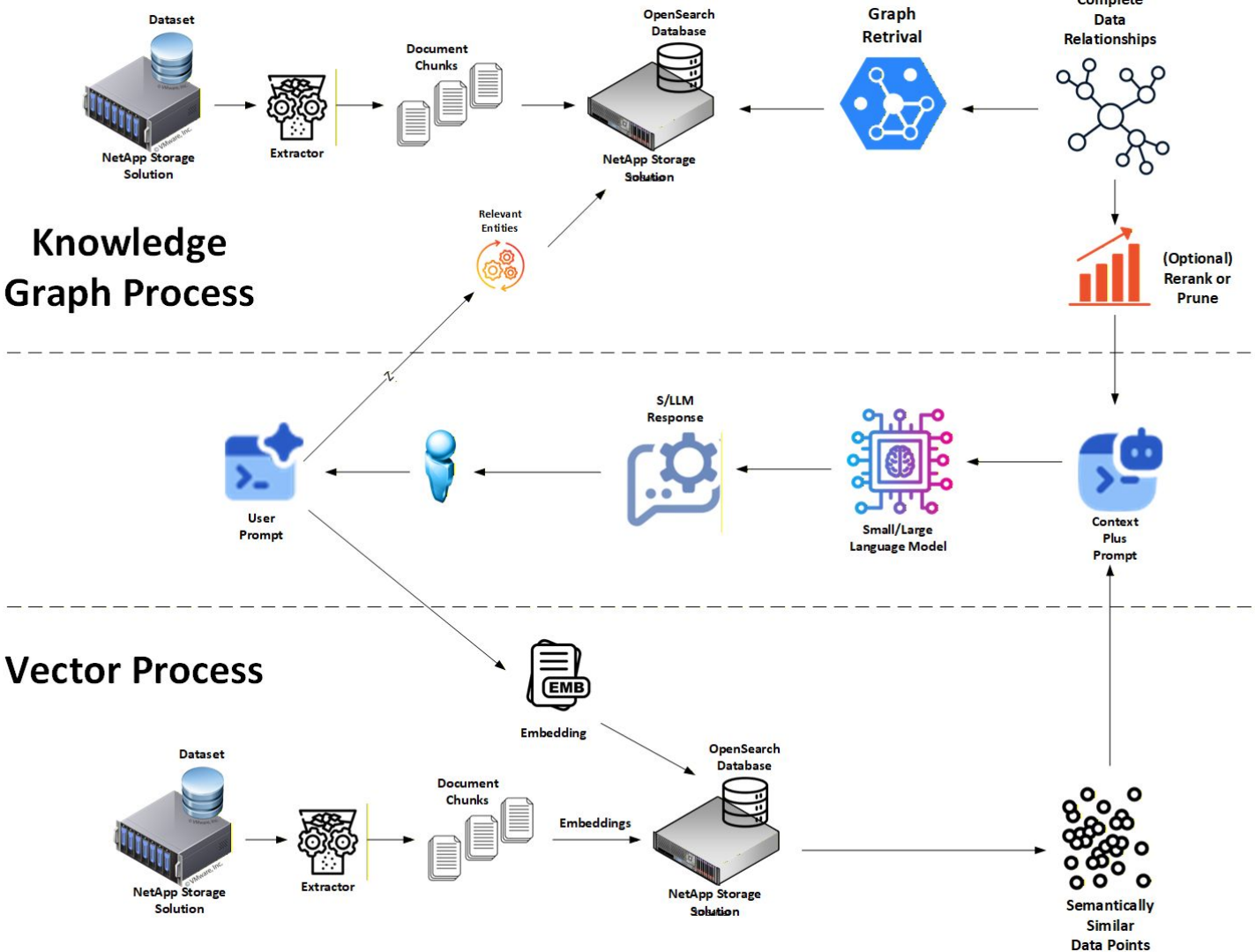


Image Attribution
A David's Original

Hybrid RAG Using Graph & Vector

- In My Opinion, Gold Standard for Factual Answer
  - Research: 96% Factual Faithfulness
  - Finance Setting Where Wrong Answers Mean 💰💰
  - By NVIDIA & Blackrock
- Near-Zero Hallucinations
- Graph Adds Complexity
  - Manage Ontologies
  - Maintenance
  - Needs Expertise
  - Infrastructure

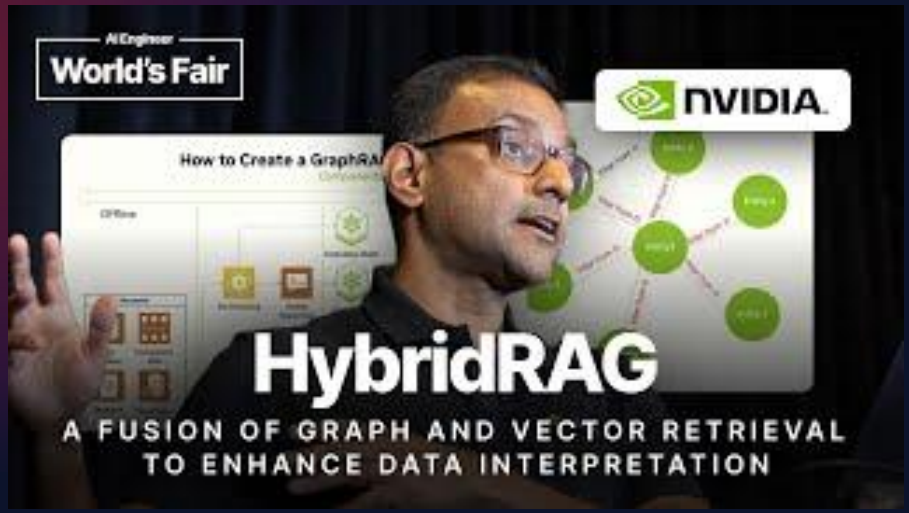


Image Attribution
Mitesh Patel (NVIDIA) at AI Engineer 2025
HybridRAG: A Fusion of Graph and Vector Retrieval

Hybrid RAG

BM25 and Vector

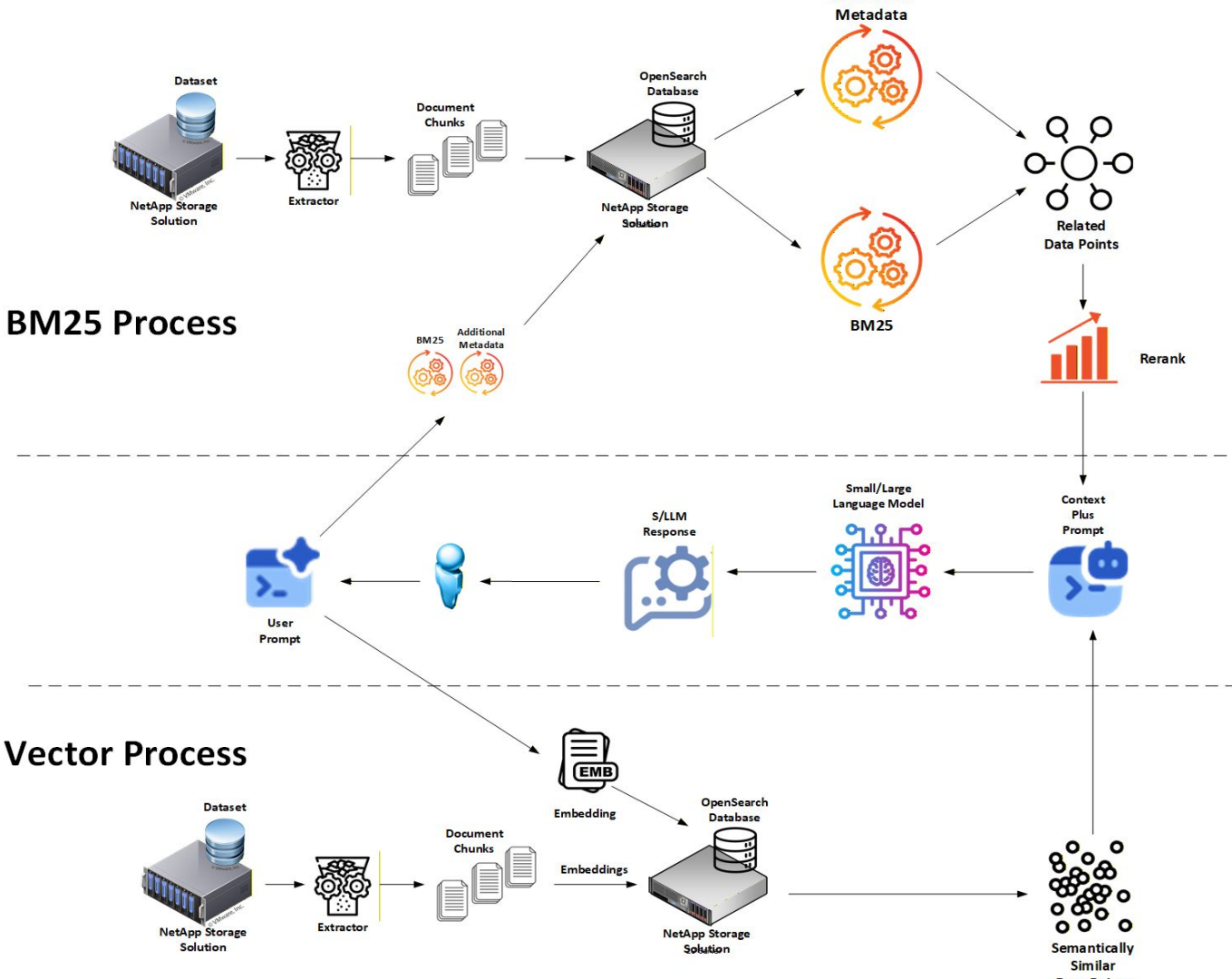


Image Attribution
A David's Original

Hybrid RAG Using BM25 & Vector

- The "Compromise" Solution
  - Predictability via BM25
  - Key/Value and Metadata Lookup
  - Also, Good for AI Governance
- Less "Work" Than Graph. Don't Need:
  - Graph Expertise or Licenses
  - Separate Infrastructure
- Trade Offs:
  - Expression of Data Relationships
  - Ease vs Factual Precision

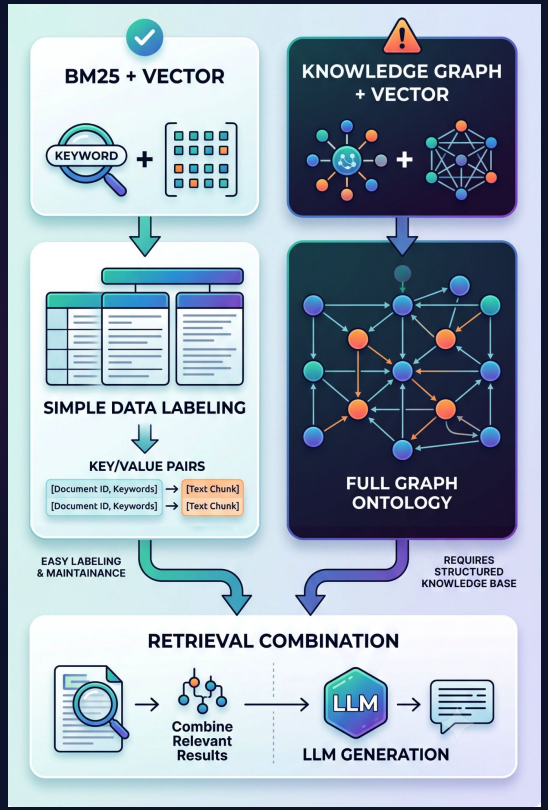


Image Attribution
Gemini Generated

Why Are AI Pilots Failing?

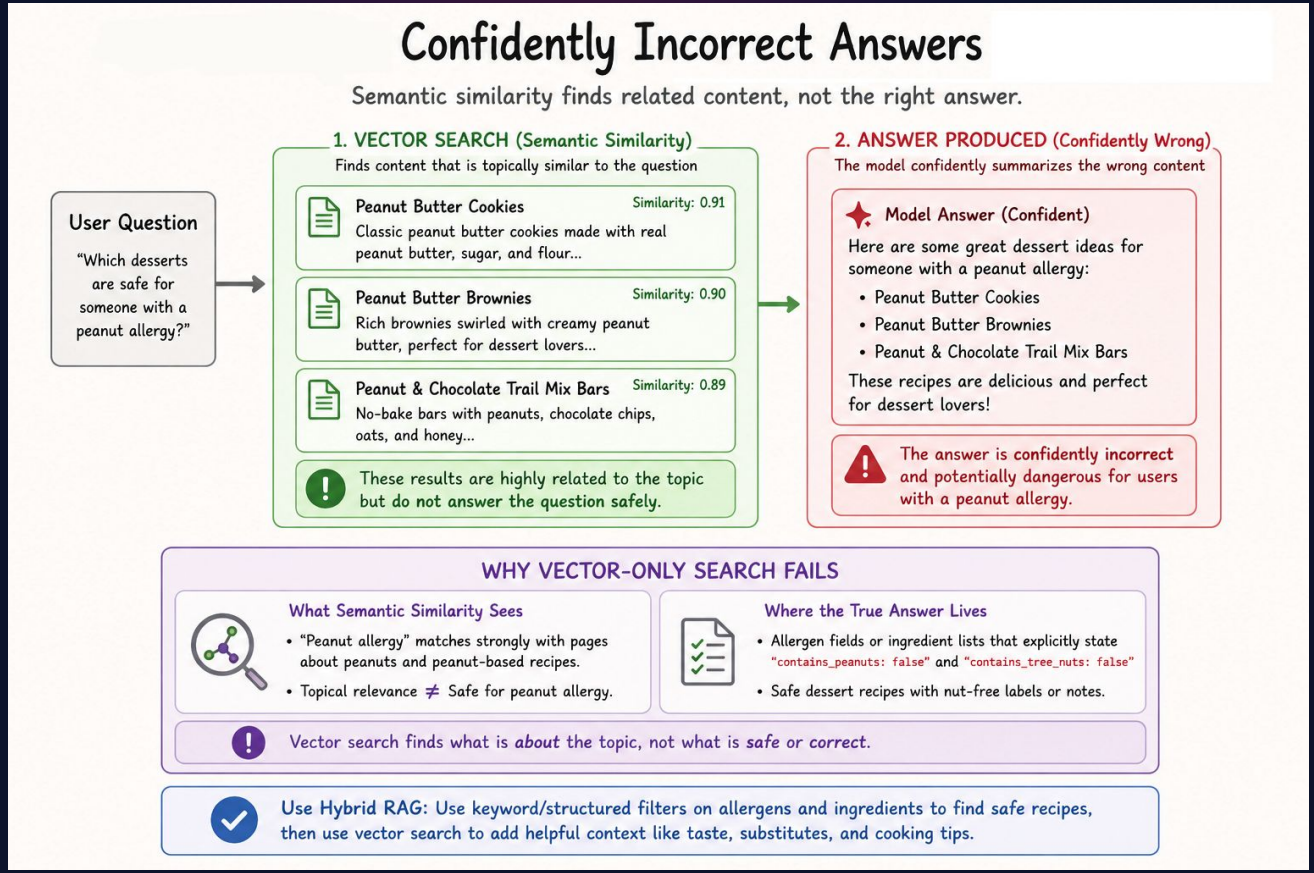
- The Realization: AI Needs Governance for Production
  - How Did My Agent Produce the Answer?
  - Why Did the LLM Provide That Answer?
  - What Data Was Used in the Answer?
  - What API/MCP Called Were Used?
- Would You Bet Your Job On:
  - Consistently Providing Same Answer
  - Ability to Debug Where $1M Went
- AI Transaction (like a SQL Transaction)
- If You Answered "No"...



Image Attribution
Princeton University Researcher - Tool helps clear biases from computer vision
Molly Sharlach on October 1, 2020

Food Allergies… Take Two!

Image Attribution
ChatGPT Generated



Section 3: How Data Affects Inference

github.com/davidvonthenen/2026-devoxx-morocco

Task 1: How Data Feeds Inference

Section 3: Results Discussion

Why the Hybrid RAG Approach?

- Anchor the Answers on Facts and Data
- A Fact or Numeric Answer Isn't Great for Conversation
- Blending Semantic and Fact is Ideal for Humans!

Takeaways:

- Retrieve The Fact and Build Around It
- AI Governance:
  - Is the Answer Observable, Repeatable, Correct, etc?
  - What "Data" Composed This Answer?
- You Want an AI "Transaction" of Data (like a SQL Transaction)

Image Attribution
Princeton University Researcher - Tool helps clear biases from computer vision
Molly Sharlach on October 1, 2020

Fact-based Answers Wrapped in AI Governance

Believe It Or Not... Harness Code

After the Claude Code Source Leak...

- We Learned the Magic is in "Harness Code"
- Belief: LLM Did ALL the Heavy Lifting
- Reality: It Was Harness Code!!
  - 98.4% of the Total Codebase



arXiv Search Submit Donate Log in

Computer Science > Software Engineering

[Submitted on 14 Apr 2026 (v1), last revised 2 Jul 2026 (this version, v2)]

Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems

Jiacheng Liu, Xiaohan Zhao, Xinyi Shang, Zhiqiang Shen

Claude Code is an agentic coding tool that can run shell commands, edit files, and call external services on behalf of the user. This study describes its architecture by analyzing

Access Paper:
View PDF
HTML (experimental)
TeX Source
(cc) BY-NC-SA view license

Current browse context:
cs.SE
< prev | next >
new | recent | 2026-04

Image Attribution
Dive into Claude Code: The Design Space of Today's and Future AI Agent Systems
https://arxiv.org/abs/2604.14228

Harness: Negation for Allergens

Mapping Attribution and Negation

- Using NER with a Negation Modifier
- https://github.com/jenojp/negspacy

Examples:

- "without soba" -> –notquery
- "noodle" -> –query

This is just Harness Code!

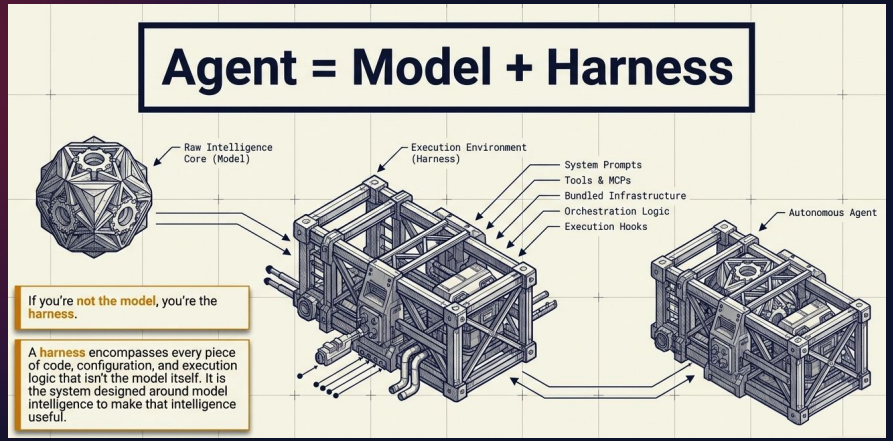


Image Attribution
What is Harness Engineering? The Three Pillars of Harness Engineering
https://medium.com/@tahirbalarabe2/what-is-harness-engineering-the-three-pillars-of-harness-engineering-1ca01f47275f

The "Final Exam" Application

- Complete the Recipe / Cooking Chatbot Application
  - Search Recipes Based Likes and Dislikes
  - Ask About Recipe Ingredients
  - Exclude Allergens (Soy, Dairy, etc)
    - High Stakes: I Will Die If I Eat a Peanut
  - Ask Questions On Preparing the Recipe
- Simple Chatbot, Serious Consequences



Image Attribution
Gemini Generated

Section 4: The "Complete" Example

github.com/davidvonthenen/2026-devoxx-morocco

Task 1: Answers with Governance

Section 4: Results Discussion

Building Out a "Simple" Chatbot:

- Need to Handle "Negation" (Default is Attribution)
  - Language is Structure, Not Math
- Structure Provides Exact Answers
  - Knowledge Graphs, SQL, Key/Value Pairs

Takeaways:

- Simple Chatbot Provides "Simple" (or Wrong) Answers
  - Even a (Production) Chatbot is Complex
- Knowledge About Your Data is IMPORTANT
- Don't Buy Into the Handwavy Simplicity

Image Attribution
Princeton University Researcher - Tool helps clear biases from computer vision
Molly Sharlach on October 1, 2020

Resources

Resources

All Materials (Slides, Code, etc):
github.com/davidvonthenen/2026-devoxx-morocco

Hybrid RAG Implementations:

- Graph (Neo4j) + Vector (OpenSearch)
- BM25 + Vector (OpenSearch for both)

YouTube: Variations In RAG Implementations

Blog: Hybrid RAG in the Real World: Graphs, BM25, and the End of Black-Box Retrieval



DEVOXX MOROCCO

Thank You!



David vonThenen
Senior AI/ML Engineer
@davidvonthenen